

Send to enrico.prati@unimi.it and daniele.bajoni@unipv.it and wait for approval. We recommend to ask approval with sufficient advance (at least 3 weeks) before the examination day you plan, so you may start to work on the project once you received approval.

Foundations of Quantum Computing 2025/26

PROPOSAL OF EXAMINATION PROJECT

Title: Quantum Phase Transition of the Transverse-Field Ising Model: a VQE and QAOA Study

Learning method/algorithm used: VQE and QAOA with a Hamiltonian Variational Ansatz (HVA); gradient-variance analysis to study trainability (barren plateaus).

Objective (max 3 lines):

Probing the QPT of the 1D TFIM using VQE and QAOA, benchmarked against exact ground states from the PennyLane qspin dataset. The critical field h_c is estimated via finite-size scaling of the magnetisation and entanglement entropy. Energy error and gradient variance vs h reveal where the variational ansatz fails and connect this to barren plateaus.

Based on a previously existing project:

- ☐ No, it is totally new not based on code of others
- ☒ Yes, I have partly used code from the following repositories (Qiskit, Mitiq, Github...):
 - **PennyLane** – statevector simulation and optimisation of the VQE and QAOA circuits.
 - **NumPy / SciPy / Matplotlib** – base numerical and plotting stack.

Reference used:

- Kadowaki & Nishimori – Quantum annealing in the transverse Ising model, Phys. Rev. E 58 (1998)
- Peruzzo et al. – A variational eigenvalue solver on a photonic quantum processor, Nat. Commun. 5, 4213 (2014)
- Farhi et al. – A Quantum Approximate Optimization Algorithm (2014)
- Calabrese & Cardy – Entanglement entropy and quantum field theory, J. Stat. Mech. P06002 (2004)
- McClean et al. – Barren plateaus in quantum neural network training landscapes, Nat. Commun. 9, 4812 (2018)
- Wiersema et al. – Exploring entanglement and optimization within the Hamiltonian Variational Ansatz, PRX Quantum (2020)
- Preskill – Quantum Computing in the NISQ Era and Beyond, Quantum 2, 79 (2018)

Needed dataset:

- ☐ Not needed
- ☐ Yes: I have original data from...
- ☒ Yes: Published here url: <https://pennylane.ai/datasets/transverse-field-ising-model>